Lifeguarding · Public Health

JLU-SPH

2026 Xinghua Cultural Festival Synthetic Biology Science Communication Booth Proposal

Safeguarding Life with Scientific Integrity

Responding to Public Health Challenges through Collaborative Efforts



School of Public Health, Jilin University

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I. Basic Information on the Event

- **Event Name:** Xinghua Cultural Festival Synthetic Biology Science Popularization Booth
- **Event Theme:** Leveraging the public, cultural, and social setting of the Xinghua Cultural Festival to conduct science popularization activities on synthetic biology for a diverse and mixed audience, objectively showcasing the technological potential and real-world limitations, and facilitating multi-level public scientific dialogue.
- **Event Type:** ☑ Public Service ☑ Science Education and Popularization ☐ Cultural Promotion ☐ Community Service ☐ Others:
- **Organizer:** JLU-SPH Team

II. Purpose of the Event

1. **Core Purpose:** Utilizing the culturally rich, socially interactive, and openly communicative public space of the Xinghua Cultural Festival to popularize synthetic biology among a diverse public; employing intuitive displays, streamlined Q&A, and contextualized explanations to guide the public in relating biotechnology to their everyday lives. Objectively introduce the potential of synthetic biology, realistically articulate the gap between research/scientific validation and real-world application, and establish public engagement as a critical component of project outreach.

2. **Specific Objectives:**

- Knowledge/Cognitive Aspect: Facilitate audiences of various ages and educational backgrounds in establishing foundational knowledge of synthetic biology ; promote awareness of both the technological prospects and real-world boundaries of research translation.
- Behavior/Skill Aspect: Engage in one-on-one communication with college students, family groups, the elderly, and general citizens to gather pressing health concerns and technical questions, and to differentiate the science education needs of various audiences.
- Emotional/Attitudinal Aspect: Lower the comprehension barrier imposed by technical terminology, enabling all audience members to easily participate in scientific conversations and to foster a rational perspective on cutting-edge biotechnology.

3. **Expected Outcomes:**

- Coverage: Nearly one hundred faculty, students, and citizens participating in the Xinghua Cultural Festival
- Completion of comprehensive activity process documentation and effectiveness feedback.

Ⅲ、Target Audience

- Primary Audience: The diverse attending public at the Xinghua Cultural Festival, including current university students, family groups, the elderly, and the general citizenry.
- Characteristics of the target population: The age range is wide and there are significant differences in knowledge background; the primary purpose is leisure tourism, and participation is often incidental; different groups have varying focuses, necessitating stratified and adaptive explanatory language. It is appropriate to utilize practical, life-based demonstration materials to facilitate the reception of scientific popularization information.

IV. Event Time and Venue

- Event Date: April 19, 2026
- Total Duration: Throughout the duration of the Xinghua Cultural Festival
- Event Venue: Nanling Campus, Jilin University

V. Event Content and Process

[Overview of Event Process]

Stage	Time	Session Title	Specific Content	Responsible Personnel	Required Materials
Stage	30 minutes prior to the event	On-site Setup and Staff Training	Set up the science popularization booth, arrange everyday-life illustration displays and portable educational materials, and complete the placement of exhibition materials	General Coordinator, Logistics Staff	Science Popularization Manuals, Promotional Posters
Main Stage	During Xinghua Festival activities	Booth Reception and Tiered Science Communication	Receive and engage with the diverse passing audience using exhibition materials for one-on-one interactions; employ tailored communication strategies for university students, family groups, the elderly, and general citizens in concise Q&A sessions	All team members	Science Popularization Manuals, Promotional Posters
Closing Stage	Event Wrap-Up	Booth dismantling and content organization	Dismantle the booth, inventory materials; organize questions from various audiences, perform comparative analysis of the different groups' science popularization needs, and archive all event materials	General Coordinator, Logistics Staff	Storage Tools

[Detailed Descriptions of Each Session]

Session 1: Booth Setup and Pre-Job Staff Training

- Core Content: Complete the science popularization booth setup for the cultural festival, arrange illustrated posters and portable educational materials with everyday themes. Conduct comprehensive staff training to standardize the science communication approach: language should be clear and measured, explaining both the possibilities of synthetic biology (合成生物学, synthetic biology) and objectively addressing the gap between research, experimental validation, and practical application; develop multiple communication templates tailored to family groups, the elderly, students, and general citizens.

Session 2: Booth Reception and Tiered Science Communication

- Core Content: Receive festival attendees of all backgrounds, carrying out science communication via intuitive displays, concise Q&A, and contextual explanations. Adjust communicationstrategies responsively to different visitors' backgrounds; reduce technical terminology barriers; do not overpromote, but objectively introduce technological advantages and existing limitations.

- Key Points: Respect the pace of visitors, do not force participation; use illustrations and relatable case studies for children; emphasize health-related topics for the elderly; appropriately expand technical depth for university students.

VI. Staffing and Division of Responsibilities

- General Coordinator (1 person): Overall planning for the booth, liaising with Xinghua Cultural Festival organizers, ensuring appropriate science communication, and handling on-site emergencies.
- Science Communicator (3-4 persons): Conduct one-on-one science communication with diverse audiences, adapting explanatory approaches according to the identity of each visitor, and objectively conveying both the technological prospects and current limitations.
- Question Recording Group (2 persons): Categorically record questions and points of interest raised by different population groups, and organize onsite feedback information.
- Logistics & Photography (2 persons): Responsible for booth setup, materials supply, and takedown/storage; document event materials through photography or filming without disturbing visitors.

VII. Materials Inventory

- Equipment Category:
  - Camera/High-definition smartphone
- Science Communication & Research Materials:
  - Popular Science Booklet
  - Promotional Poster
- Logistics Category:
  - Simple Tables and Chairs
  - Ballpoint Pens
  - Trash Bags

VIII. Risk Mitigation Plan

- Personnel Risk: High onsite visitor flow and simultaneous presence of multiple audience groups may result in insufficient staff for reception. Flexible allocation of personnel for reception, guiding visitors to view illustrated exhibits independently, and implementation of staggered one-on-one communications are recommended solutions.
- Venue Risk: Outdoor strong winds and rainfall. Secure display boards and posters with heavy objects and use protective coverings for materials against rain.In case of extreme weather, distribution of popular science information QR codes can replace physical displays, thereby reducing onsite activities.
- Material Risk: Printed science communication materials may be exhausted ahead of schedule. QR codes for digital versions of popular science resources can direct the public to access information online.

IX. Outcome Evaluation

- Process Evaluation: Record the number of onsite participants and the duration of their exchanges; observe engagement levels among different population groups. Sort and analyze categorized
- recording forms to compare the question patterns of family groups, elderly groups, and non-bioscience major students, in order to assess the science communication needs specific to each group.
- Satisfaction Evaluation: Conduct brief oral interviews with voluntary participants to gather feedback on their experiences with the science communication content.
- Review and Archiving: Compile questions and needs from various population groups, summarize directions for optimizing science communication materials, and archive textual and visual documentation of the event. This will provide practical references for future science communication activities targeting mixed audiences.